



# Higher Technical School of Telecommunications Engineering and Systems



## RADIOCOMMUNICATION SYSTEMS

### PRACTICE 2

### DVB-T REPEATER PLANNING

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# Contents

## Contents - 2

- 1 Introduction - 3
- 2 General description of the repeater - 4
- 3 Selected site and coverage objective - 5
  - 3.1. Location - 5
  - 3.2. Profiles - 6
  - 3.3. Electrical power supply - 7
  - 3.4. Repeater-station simulation - 8
- 1 Selected equipment - 11
- 2 Conclusion - 13
- 3 Appendix - 14

# 1. Introduction

Digital terrestrial television (DTT) repeaters, commonly called gap fillers, are essential elements for extending terrestrial digital television coverage. They operate on a receive-and-retransmit principle: the signal from a main transmitter is captured at an elevated site and retransmitted toward urban areas that do not have direct coverage. The Spanish regulatory framework established by Royal Decree 944/2005 limits the apparent radiated power to 1 W in order to preserve electromagnetic compatibility with other services.

In this practice, the repeater is designed to provide DTT service to Reduena, a small municipality in the foothills of the Sierra de Guadarrama. The settlement is located in an enclosed-valley area surrounded by extensive agricultural land.

Under the national DTT technical plan, Spain uses the UHF frequency band from 470 to 790 MHz with 8 MHz channels and the 8K operating mode specified by the European EN 300 744 standard. The 8K mode uses 6,817 total carriers, of which 6,048 carry data. The proposed repeater operates at 670 MHz, corresponding to channel 47.

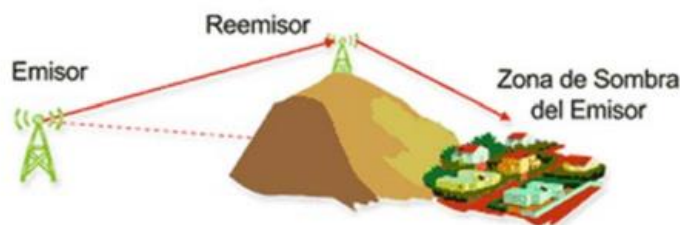


Figura 1. Esquema de funcionamiento de un reemisor

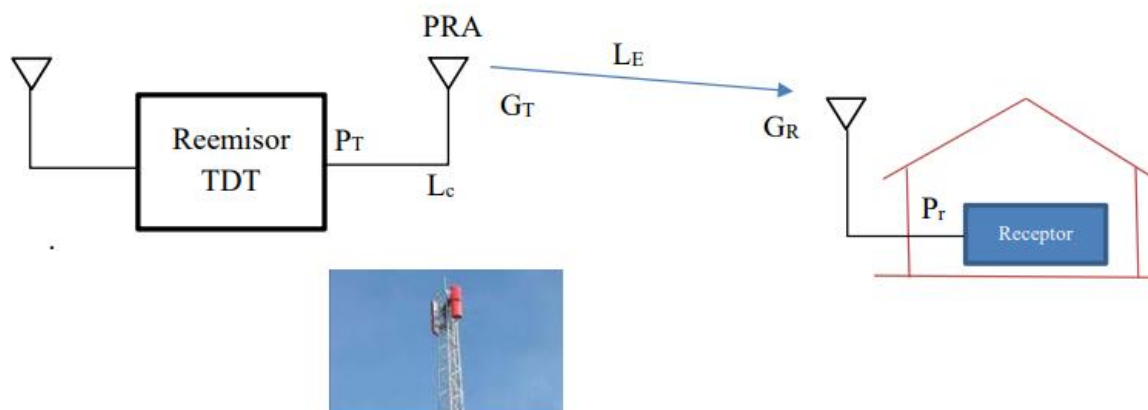


Figure from source page 3.



Figure from source page 3.

## 2. General Description of the Repeater

The repeater designed for Reduena operates under the DVB-T standard with an 8K, 64-QAM and FEC 2/3 configuration, providing a useful bit rate of 19.91 Mbps. These parameters ensure compatibility with the national DTT network while making efficient use of the spectrum.

The system follows the conventional receive-process-transmit architecture. Its receiving stage captures the signal from the main transmitter at Torrelaguna. The signal is then frequency-converted, processed and amplified before being retransmitted toward Reduena.

### Repeater parameters

The transmitter and domestic-receiver parameters are shown in the original tables.

### Important: Domestic TV receiver parameters

The receiver antenna gain, feeder losses and reception threshold are included in the original figure.

### Objective of the Reduena repeater stages

The system must provide a configuration that optimally regenerates the DVB-T signal from the Torrelaguna main transmitter and delivers it to the population of Reduena.

Reception architecture -> central processing -> transmission stage.



Figure from source page 4.

| RECEPTOR TV DOMÉSTICO |         |
|-----------------------|---------|
| Ganancia antena       | 14 dB   |
| Perdidas del feeder   | 5 dB    |
| Umbral                | -65 dBm |

Figure from source page 4.

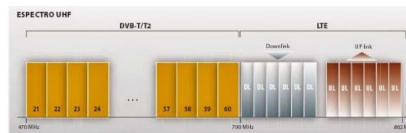


Figure from source page 4.

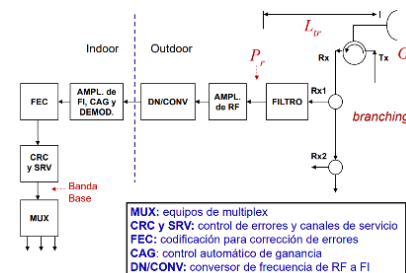


Figure from source page 4.

| TRANSMISOR DEL REEMISOR                      |        |
|----------------------------------------------|--------|
| Ganancia antena                              | 14dB   |
| Potencia                                     | 23dBm  |
| Perdidas del alimentador de antena( feeder ) | 7 dB   |
| PRA máxima                                   | 30 dBm |

Figure from source page 4.

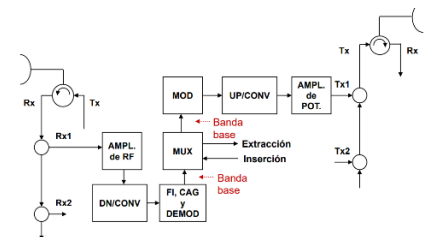


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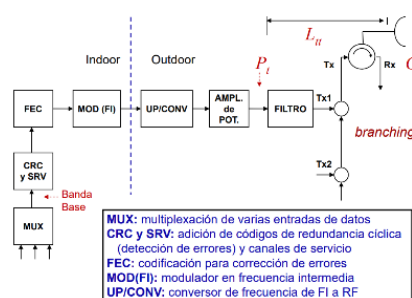


Figure from source page 4.

## 3. Selected Site and Coverage Objective

### 3.1. Location

Reduena, postal code 28721, is located at approximately 40 deg 49 min 21.43 sec N, 3 deg 35 min 59.00 sec W in the Community of Madrid. Its population is below 1,000 inhabitants, so it meets the practice criterion.

An analysis of the available digital multiplexes identified five channel groups already operating on channels 25, 32, 33, 34 and 38. Channel 47 at 670 MHz was therefore selected to reduce the risk of interference while retaining the favorable propagation characteristics of the UHF band.

The basic siting criteria were:

- 1 The repeater must be less than 20 km from a nearby DTT transmitter.
- 2 The repeater must be less than 5 km from the settlement to be served.



**Código Postal: 28721 Población: REDUEÑA**

Un múltiple digital es un conjunto de canales de televisión. En cada localidad, cada múltiple digital se emite por una frecuencia diferente.

La tabla que se ofrece a continuación muestra las frecuencias por las que se emite cada múltiple digital y el nombre del centro emisor de señales de televisión que cubre el código postal consultado.

Tu localidad puede recibir las señales de TDT por más de un centro emisor, por lo que podría recibir los canales de televisión por más de una frecuencia. Si este es el caso, se mostrará en una tabla adicional hasta dos centros emisores desde los que, en ese código postal, se reciben los canales de TDT.

| Múltiple digital | Centro  | Canal |
|------------------|---------|-------|
| RGE              | REDUEÑA | 33    |
| MPE1             | REDUEÑA | 32    |
| MPE2             | REDUEÑA | 34    |
| MPE3             | REDUEÑA | 25    |
| MAUT             | REDUEÑA | 38    |

Figure from source page 5.



**Canales de televisión en cada múltiple digital**

La imagen que se ofrece a continuación muestra los canales de televisión que se emiten en los múltiples digitales nacionales y autonómicos.

**Organización de los canales de TDT**

Legend:

- RGE: Red Global Estatal (red que emite canales de televisión públicos de ámbito estatal)
- MPE: Múltiple Privado Estatal (red que emite canales privados de televisión de ámbito estatal)
- MAUT: Múltiple Autonómico (red que emite canales públicos y privados de televisión de ámbito autonómico)

Figure from source page 5.



Figure from source page 5.





Figure from source page 5.

### 3.2. Profiles

The terrain profiles show adequate clearance on the two critical radio paths. The path from the Torrelaguna transmitter to the repeater is approximately 5 km long and has controlled elevation differences. The path from the repeater to Reduena is approximately 800 m long and has excellent line of sight.

Clearance of the first Fresnel zone is greater than 60% at the critical points, which provides favorable propagation conditions.

#### Transmitter-to-repeater profile

The original figure shows the terrain profile and Fresnel-zone clearance.

#### Settlement-to-repeater profile

The original figure confirms the short, unobstructed path to the target population.

$$L_b = \underbrace{L_{bf} + L_g}_{\text{pérdidas para condiciones normales}} + L_d + L_{desv} + L_p$$

$L_{bf}$  : pérdidas de espacio libre  
 $L_g$  : atenuación por gases atmosféricos  
 $L_d$  : pérdidas por difracción; despejamiento insuficiente en condiciones adversas de propagación  
 $L_{desv}$  : pérdidas por desvanecimiento (multitrayecto, no precipitación)  
 $L_p$  : atenuación debida a precipitación

Figure from source page 6.

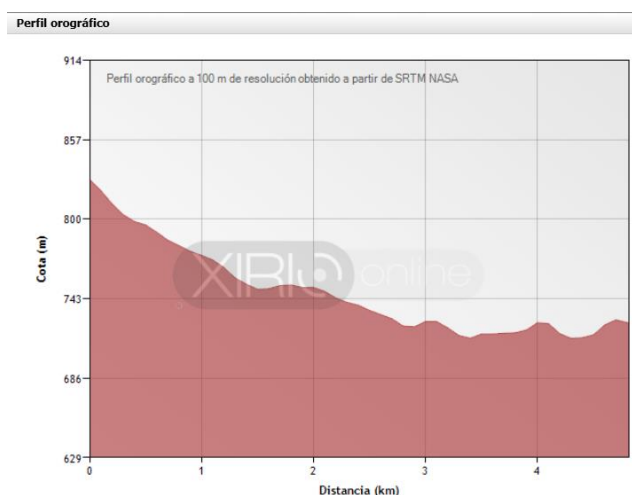


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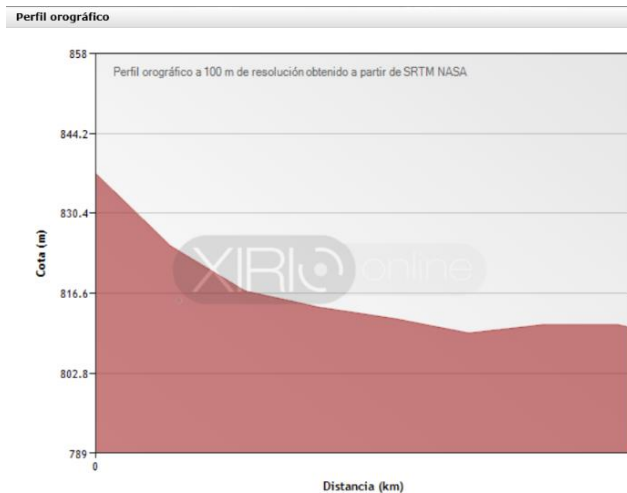


Figure from source page 6.

### 3.3. Electrical Power Supply

A suitable power-supply point is available close to the selected site. This short connection distance has a positive effect on both installation and maintenance.

#### 1. Extremely low feeder losses

- AC cable losses are negligible, below approximately 0.1%.
- The voltage remains stable without significant voltage drop.
- A minimum practical conductor cross-section, such as 1.5 mm<sup>2</sup>, can be used.

#### 2. Reduced installation cost

- Only about 250 m of cable is required instead of several kilometers.
- Less extensive surge protection is needed.
- Installation can be completed in hours rather than days.

#### 3. Improved reliability

- The short route contains fewer possible failure points.
- Maintenance and fault location are simpler.
- The cable is less exposed to external damage.

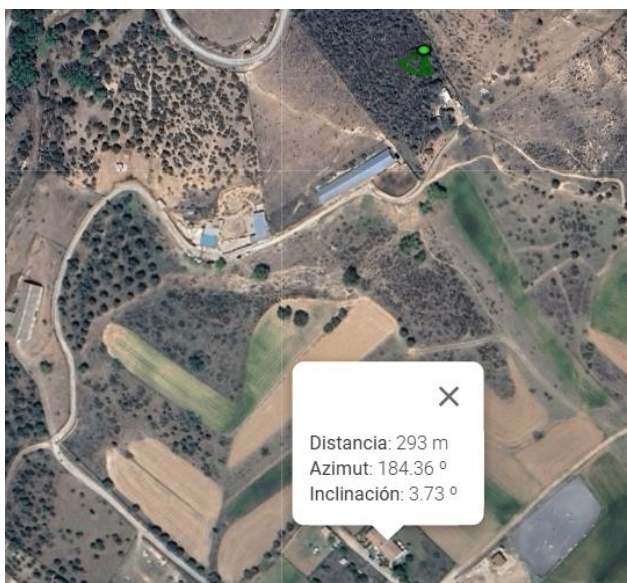


Figure from source page 7.

### 3.4. Repeater-Station Simulation

The station was simulated in Xirio using the ITU-R P.526-15 propagation recommendation, which was considered the most suitable model for this case.

The Okumura-Hata model was also tested, but its empirical assumptions are mainly oriented toward urban environments. They do not describe the very low-density, almost rural layout of Reduena particularly well. With that model it was difficult to demonstrate the required 90% population coverage with a 10 dB margin above the reception threshold.

- Utilización: enlaces terrenales punto a zona (radiodifusión, servicios móvil terrestre y marítimo) y punto a multipunto
- Colección de **curvas normalizadas** ( $p_{ra} = 1$  kW) de **campo eléctrico**
  - ✓ Distancias 1-1000 km
  - ✓ Tierra, mares cálidos y mares fríos
  - ✓ Alturas  $h_1$  de transmisor. Valores normalizados entre 10 y 1200 m
  - ✓ Altura de antena receptora  $h_2 = 10$  m
  - ✓ Frecuencias nominales: 100, 600 y 2000 MHz. Extrapolable a cualquier frecuencia entre 30 y 3000 MHz
  - ✓ Valores rebasados en el 50% de las ubicaciones, para el 1%, 10% y 50% del tiempo

Figure from source page 8.



**Método de cálculo**

**Nombre:** Nuevo Okumura-Hata Modificado

**Método de propagación:** Okumura-Hata Modificado

☐ Método híbrido basado en Okumura-Hata corregido por las difracción en obstáculos, válido en la gama 150 MHz a 2 GHz. Recomendado para servicios de móviles y de acceso de banda ancha en entornos urbanos cuando se disponga de cartografía de alta resolución.

**Resolución:** 100 m/pixel

☐ Limitar la distancia máxima del cálculo

**Límite:** 0 m

**Parámetros básicos**

☐ Cálculo automático de factor K

**Factor K:** 1.333

☐ Usar margen de desvanecimiento desde clutter

**Margen de desvanecimiento:** 0 dB

**Propiedades de Okumura-Hata Modificado**

**Medio:** Urbano

**Tipo ciudad:** Pequeña

**Penetración en edificios**

☒ No aplicar

☐ No calcular en edificios

☐ Calcular sólo en azoteas

☐ Pérdidas fijas en edificios

Figure from source page 8.

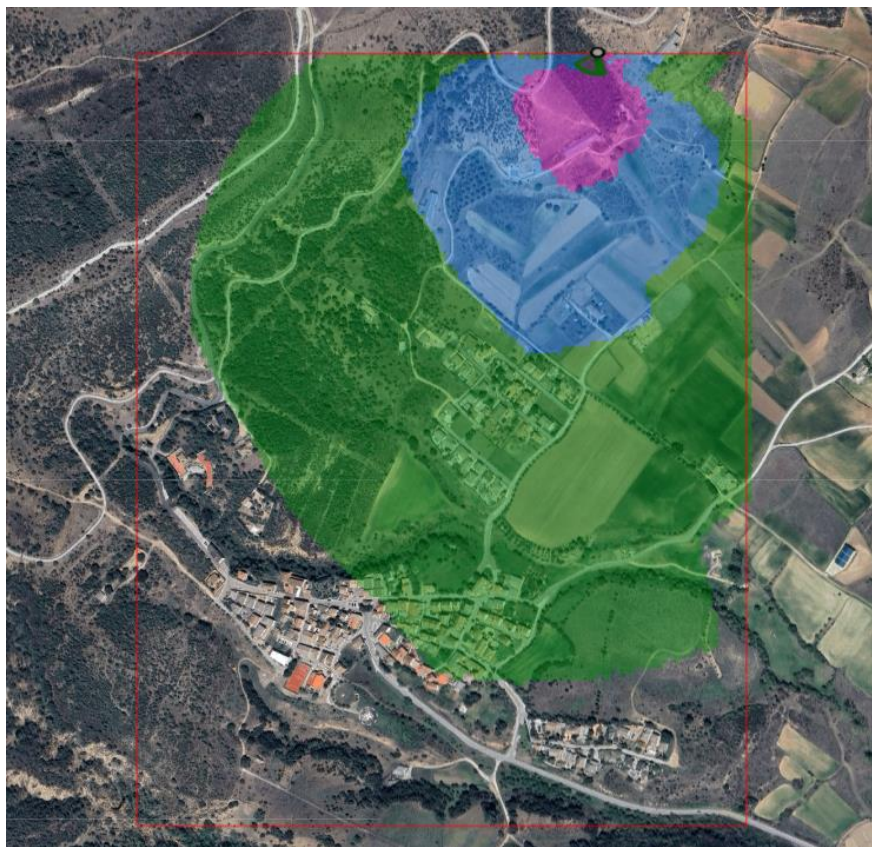


Figure from source page 8.

## Coverage Results

The final configuration concentrates coverage on Reduena and was evaluated at the -65, -55 and -45 dBm levels.

- Basic coverage above -65 dBm reaches approximately 0.73 km<sup>2</sup>, or 99.88% of the analyzed settlement area.
- Coverage above -55 dBm reaches approximately 0.64 km<sup>2</sup>, or 87.31%, and provides the required 10 dB margin over the basic threshold for most of the population.
- Excellent coverage above -45 dBm reaches approximately 0.07 km<sup>2</sup>, or 9.48%.

The central area receives levels between approximately -55 and -45 dBm, providing robust reception under multipath and weather variations. The peripheral area remains above -65 dBm, which preserves continuous basic service.



Figure from source page 9.

| Topónimo | Total S km <sup>2</sup> | S ≥ -65dBm | S ≥ -55dBm | S ≥ -45dBm | %S ≥ -65dBm | %S ≥ -55dBm | %S ≥ -45dBm |
|----------|-------------------------|------------|------------|------------|-------------|-------------|-------------|
|          | 0.73                    | 0.73       | 0.64       | 0.07       | 99.88%      | 87.31%      | 9.48%       |

Figure from source page 9.

## Coverage Including the Surrounding Area

The extended calculation was also evaluated at -65, -55 and -45 dBm. The neighboring settlement of Cotos de Monterey receives only marginal coverage near -65 dBm.

Potential benefits include:

- Additional social value through free DTT access.
- A reduction in the digital divide and improved access to public information services.
- No additional infrastructure cost for the neighboring area.

Potential drawbacks include:

- Unwanted overlap with other transmitters and possible interference.
- Equalization problems when several signals are received simultaneously.
- Incorrect automatic selection of the preferred signal in some receivers.

An azimuth of 195 degrees, a mechanical tilt of 4 degrees and an antenna height of 10 m concentrate the transmitted energy on the populated area and minimize radiation toward other directions. The result validates both the site and the final antenna configuration.

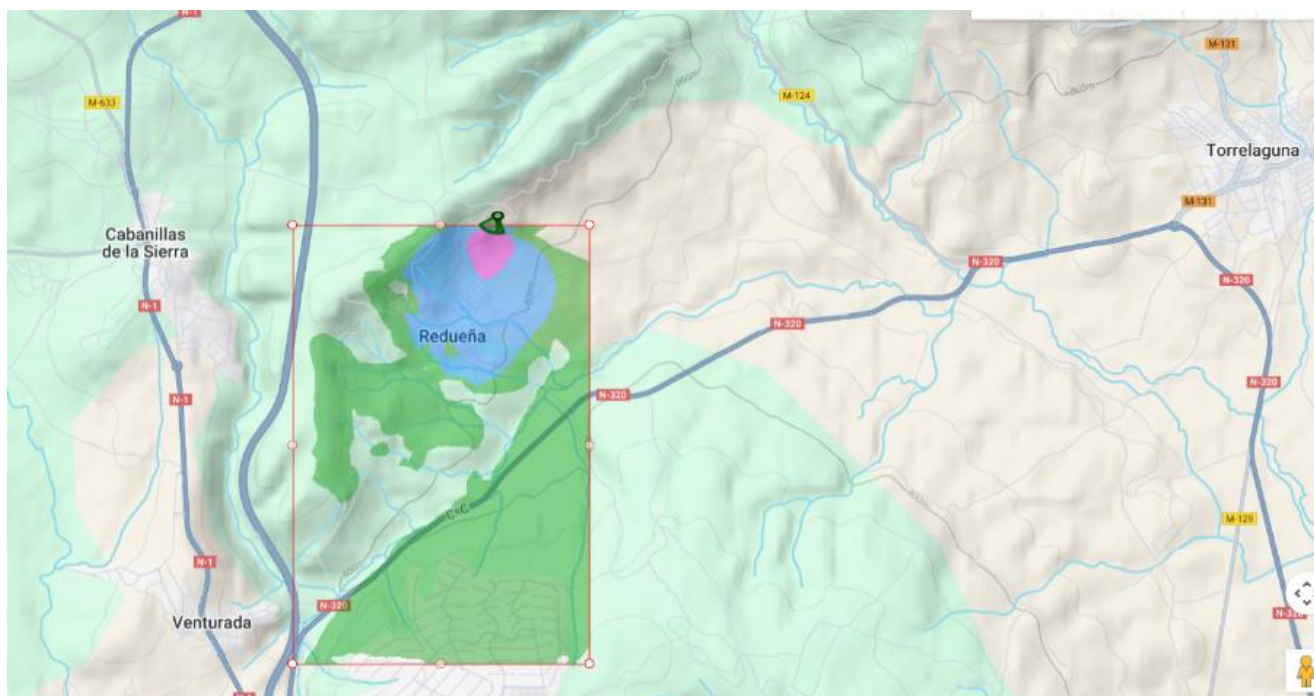


Figure from source page 10.

## 4. Selected Equipment

### Tower

A 10 m Atenea Host tower was selected because it matches the height used in the simulation. Its hot-dip-galvanized construction is suitable for long-term rural installation, and its modular design reduces installation and maintenance costs.

### Radio System

The TRedess MultiTX system was selected because it is specifically designed for gap-filler applications. Its approximately 35 W consumption is more efficient than a generic 60 W alternative. Remote monitoring reduces operating and maintenance costs, while interference rejection above 70 dB supports service quality in a congested spectrum environment.

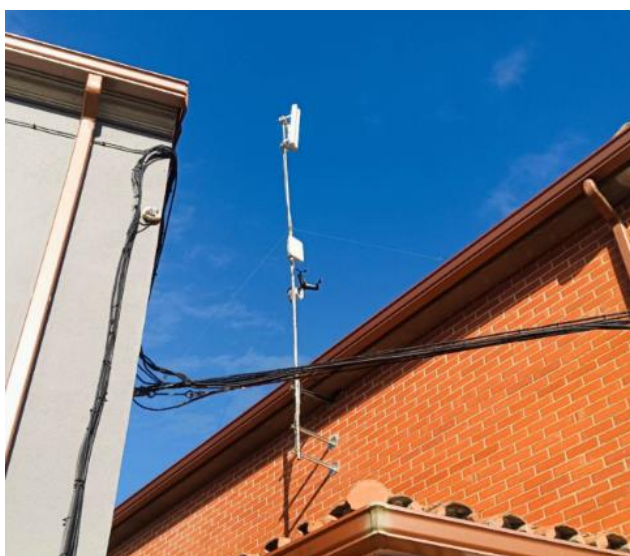


Figure from source page 11.



Figure from source page 11.





Figure from source page 11.

## Antenna

The OMB PD-2000 antenna was selected because it provides approximately 14 dBi gain, a UHF point-to-multipoint radiation pattern and a front-to-back ratio above 25 dB. Its robust mechanical design is appropriate for the installation.

The additional gain provides roughly 2 dB of extra link margin, improving reliability near the coverage boundary. Support for horizontal or vertical polarization also provides flexibility for future DTT-network changes.

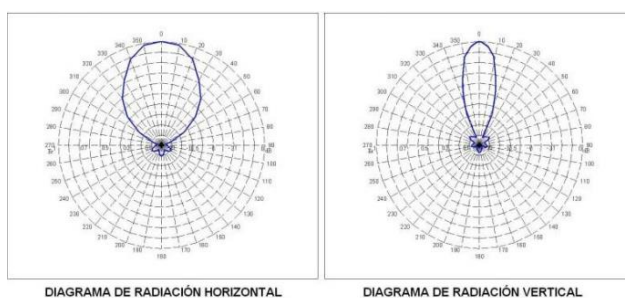


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## 5. Conclusion

The proposed DTT repeater is technically feasible and economically reasonable. The simulation provides 99.88% coverage above -65 dBm and exceeds the requirement to serve at least 90% of the population with a 10 dB margin in the principal inhabited area.

The selected coordinates are favorable, and both radio paths maintain more than 60% first-Fresnel-zone clearance. The combination of the TRedess MultiTX, the 14 dBi OMB PD-2000 antenna and the 10 m Atenea tower provides a balanced solution in terms of performance, durability and efficiency.

The power source is only about 230 m from the site, which substantially reduces the cost compared with a remote electrical installation. The ITU-R P.526-15 simulation confirms excellent coverage of the target settlement and only limited, incidental extension toward Cotos de Monterey.

The practice also developed independent technical research, optimization, standards analysis and teamwork, and provided a practical understanding of how a DTT gap filler extends the service of a main transmitter.

## 6. Appendix

### Radio-system data sheet

The original appendix contains the manufacturer's technical documentation and reference links for the selected radio system.

|                                   | 1W                                                                                             | 5W           | 10W         |
|-----------------------------------|------------------------------------------------------------------------------------------------|--------------|-------------|
| Max. Output Power (before filter) | 1.58W (32dBm)                                                                                  | 7.5W (39dBm) | 14W (41dBm) |
| Frequency range                   | 470-698 MHz                                                                                    |              |             |
| Modulation Standards              | DVB-T/T2, ISDB-T/Tb, ATSC                                                                      |              |             |
| TRANSMITTERS                      |                                                                                                |              |             |
| Inputs                            | 2xASI BNC female + 2xTSOP 10/100/1000 RJ45                                                     |              |             |
| MER                               | >35dB (Typical 42dB at nominal output power)                                                   |              |             |
| Shoulders                         | >38dB                                                                                          |              |             |
| Pre-correction                    | Digital adaptive                                                                               |              |             |
| RF output connector               | N female                                                                                       |              |             |
| 10MHz Reference input             | BNC female 50 Ω (-20dBm a +10dBm)                                                              |              |             |
| 1pps Reference input              | BNC female (TTL)                                                                               |              |             |
| GAP FILLERS                       |                                                                                                |              |             |
| Input RF level                    | -70 to -20dBm                                                                                  |              |             |
| MER                               | >34dB for gain margin (echo - signal) = +10dB<br>>29dB for gain margin (echo - signal) = +20dB |              |             |
| Echo canceller                    | Gain margin (echo - signal) up to +30dB. Flexible cancellation windows<br>Doppler cancellation |              |             |
| Shoulders                         | >38dB                                                                                          |              |             |
| Pre-correction                    | Digital adaptive                                                                               |              |             |
| RF output connector               | N female                                                                                       |              |             |
| MANAGEMENT                        |                                                                                                |              |             |
| Ethernet Capabilities             | Gigabit switch ethernet 4 ports                                                                |              |             |
| WEB server                        | HTML 5, responsive design                                                                      |              |             |
| Protocols                         | IPv4, IPv6, DHCP, FTP, SSH, NTP<br>HTTP, HTTPS, SNMP v1/v2c<br>IPSEC, PPP, PPPoE               |              |             |
| Modem                             | LTE CAT M1/M4 (antenna and modem integrated) (optional)                                        |              |             |
| Battery                           | Ion-lithium (optional)                                                                         |              |             |
| Monitoring receiver               | Parameters: QoS MER, BER (DVB-T/T2 RF channels 8MHz UHF)                                       |              |             |
| I/O Contacts                      | Inputs: 5 Optocoupled/Outputs: 5 contact relays free potential                                 |              |             |
| USB                               | USB 2.0 compatible                                                                             |              |             |
| COMMON FEATURES                   |                                                                                                |              |             |
| AC Power supply                   | AC Input & Redundant power supply as option                                                    |              |             |
| Input AC voltage                  | 85 Vac to 275 Vac                                                                              |              |             |
| Power factor                      | 0.99                                                                                           |              |             |
| Local Control                     | LCD screen and buttons (external) / Ethernet (Management Module option)                        |              |             |
| Remote Control                    | Ethernet (Web server and SNMP agent) / I/O contacts (Management Module option)                 |              |             |
| Monitoring                        | Main parameters (RF power, MER, input bit rate, internal temperature, voltage...)              |              |             |
| Temperature range                 | 0°C to 45°C                                                                                    |              |             |
| Humidity                          | < 95% @ 40°C, without condensation                                                             |              |             |
| Cooling                           | Air cooled                                                                                     |              |             |
| Subrack Dimensions (W*H*D)        | 19" x 5HU x 255mm                                                                              |              |             |
| Safety                            | EN 60950-1:2006+A1:2010+A12:2009+A12:2011<br>EN 60215:1989+A1:92+A2:94                         |              |             |
| EMC                               | ETSI EN 301 489-1 V2.2.3 / ETSI EN 301 489-53 V1.1.1                                           |              |             |
| Spectrum Efficiency               | ETSI EN 302 296 V2.2.0                                                                         |              |             |
| RoHS                              | EN 50581:2012                                                                                  |              |             |

Figure from source page 14.



## Repeater-antenna data sheet

The original appendix contains the manufacturer's technical documentation and reference links for the selected repeater antenna.

### CARACTERÍSTICAS GENERALES

|                                    |                             |
|------------------------------------|-----------------------------|
| RANGO DE FRECUENCIA                | 470-890MHz                  |
| GANANCIA                           | 11.2 dBd                    |
| 3dB BEAM WIDTH                     | Plano E: 62°, Plano H: 28°  |
| POLARIZACIÓN                       | Horizontal                  |
| IMPEDANCIA                         | 50 Ohm                      |
| VSWR                               | ≤1.10:1 (<26dB return loss) |
| MÁXIMA POTENCIA HANDLING PEAK SYNC | 2KW                         |
| MÁXIMA POTENCIA HANDLING RMS       | 1KW rms                     |
| CONECTOR                           | 7/16"(H)                    |
| DIMENSIONES                        | 482.6x203.2x965.2mm         |
| VELOCIDAD MÁXIMA DEL VIENTO        | 200km/h, 124M/h             |
| CALGA AL VIENTO (FRONTAL)          | 740NW                       |

Figure from source page 15.

|                           |              |
|---------------------------|--------------|
| CARGA AL VIENTO (LATERAL) | 260NW        |
| PESO                      | 15.8Kg       |
| ESPACIO VERTICAL          | 1000mm       |
| ATERRAMIENTO              | DC grounding |
| RANGO DE TEMPERATURA      | -30 a 70°C   |
| HUMEDAD                   | 100%         |

### SISTEMAS DISPONIBLES

| MODELO     | Nº ANTENAS | GANANCIA<br>TÍPICA (dBd) | CNC ENTRADA | POTENCIA MÁX. |
|------------|------------|--------------------------|-------------|---------------|
| PD 2000    | 1          | 11.2                     | 7/16"(H)    | 2KW           |
| PD 2000 2  | 2          | 14.1                     | N(H)        | 500W          |
| PD 2000 2R | 2          | 14.1                     | EIA 7/8"(H) | 3KW           |
| PD 2000 3  | 3          | 16                       | N(H)        | 500W          |
| PD 2000 3R | 3          | 16                       | EIA 7/8"(H) | 3KW           |
| PD 2000 4  | 4          | 17.1                     | 7/16"(H)    | 1KW           |
| PD 2000 4R | 4          | 17.1                     | EIA 7/8"(H) | 3KW           |

Figure from source page 15.